$$\mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}\tau} \equiv \mathrm{e}^{\mathrm{i}\sum_i \hat{H}_{\mathcal{E}}^{(i)}\tau} = \prod_i \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau}. \tag{5.171}$$